

CS5275 Problem Set 2

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Problem 1

a. Recall the definition of $\Phi_C(S)$ given as

$$\Phi_C(S) = \frac{|E(S, V(C) \setminus S)|}{\min(\text{vol}(S), \text{vol}(V(C) \setminus S))},$$

where $|E(S, V(C) \setminus S)|$ is precisely the number of edges removed at each step.

Denote the connected component and the subset we pick at step i as C_i and S_i , respectively. WLOG assume that $\text{vol}(S_i) \leq \text{vol}(V(C_i) \setminus S_i)$. Denote E_i to be the set of edges removed at step i . Hence by $\Phi_{C_i}(S_i) \leq \phi$, we have $|E_i| \leq \phi \text{vol}(S_i)$.

For every $v \in V$, the number of times it can possibly appear in some S_i is at most $\log_2(\text{vol}(V))$, since at each step we split some connected component into two which could at most halve the volume. Note that for $G = (V, E)$, we have $\text{vol}(V) \leq 2|E|$ and $|E| \leq \binom{n}{2}$. Thus we have

$$\log_2(\text{vol}(V)) \leq \log_2(2|E|) \leq \log_2\left(2 \cdot \binom{n}{2}\right) = \log_2(n(n-1)) \in O(\log n).$$

Therefore the total number of edges removed during this process is at most

$$\sum_i |E_i| \leq \phi \sum_i \text{vol}(S_i) \in O(\phi \cdot |E| \cdot \log n),$$

as required.

Problem 2

a. We construct $G = (V, E)$ as follows:

- Let A and B be two $\lfloor n/3 \rfloor$ -cliques.
- Let P be a path graph with $n - 2\lfloor n/3 \rfloor$ vertices, $v_0, v_1, \dots, v_m$ with $m \in \Theta(n)$.
- Connect v_0 to a vertex in A and v_m to a vertex in B .

We also define a function $f : V \rightarrow \mathbb{R}$ as follows:

$$f(v) = \begin{cases} 1, & v \in A, \\ -1, & v \in B, \\ 1 - 2i/m, & v_i \in P, \quad i = 0, 1, \dots, m. \end{cases}$$

We have $\deg(v) = \lfloor n/3 \rfloor - 1 \in \Theta(n)$ for all $v \in A$ and $v \in B$ and $\deg(v) = 2$ for all $v \in P$. Therefore

$$\begin{aligned} \sum_{v \in V} \deg(v) f(v) &= \sum_{v \in A} \deg(v) f(v) + \sum_{v \in B} \deg(v) f(v) + \sum_{v \in P} \deg(v) f(v) \\ &= \left(\sum_{v \in A} \deg(v) \cdot 1 + \sum_{v \in B} \deg(v) \cdot (-1) \right) + \sum_{i=0}^m 2 \left(1 - \frac{2i}{m} \right) \\ &= 0 + 0 = 0, \end{aligned}$$

so via the variational characterization given for λ_2 , we have

$$\lambda_2(\mathbf{N}) \leq \frac{\sum_{\{u,v\} \in E} (f(u) - f(v))^2}{\sum_{v \in V} \deg(v) f(v)^2}.$$

The denominator can be bounded as

$$\begin{aligned} \sum_{v \in V} \deg(v) f(v)^2 &= \sum_{v \in A} \deg(v) f(v)^2 + \sum_{v \in B} \deg(v) f(v)^2 + \sum_{v \in P} \deg(v) f(v)^2 \\ &= \left(\sum_{v \in A} \deg(v) \cdot 1^2 + \sum_{v \in B} \deg(v) \cdot (-1)^2 \right) + \sum_{i=0}^m 2 \left(1 - \frac{2i}{m} \right)^2 \\ &= 2 \left\lfloor \frac{n}{3} \right\rfloor \left(\left\lfloor \frac{n}{3} \right\rfloor - 1 \right) + \frac{2(m+1)(m+2)}{3m} \in \Theta(n^2). \end{aligned}$$

For the numerator, notice that for any edge $\{u, v\}$ in A and B , the quantity $(f(u) - f(v))^2 = 0$, as f is constant. Thus we only need to consider the edges in P , $\{v_0, v_1\}, \{v_1, v_2\}, \dots, \{v_{m-1}, v_m\}$, the edge $\{v_A, v_0\}$ connecting P to A and the edge $\{v_m, v_B\}$ connecting P to B .

Indeed, we have

$$\begin{aligned} \sum_{\{u,v\} \in E} (f(u) - f(v))^2 &= \sum_{i=0}^{m-1} \underbrace{\left(\left(1 - \frac{2i}{m} \right) - \left(1 - \frac{2(i+1)}{m} \right) \right)^2}_{\text{all edges in } P} \\ &\quad + \underbrace{\left(1 - \left(1 - \frac{2 \cdot 0}{m} \right) \right)^2}_{\text{the edge } \{v_A, v_0\}} + \underbrace{\left(\left(1 - \frac{2 \cdot m}{m} \right) - (-1) \right)^2}_{\text{the edge } \{v_m, v_B\}} \\ &= \frac{4}{m} + 0 + 0 \in \Theta\left(\frac{1}{n}\right). \end{aligned}$$

Putting together, we have

$$\lambda_2(\mathbf{N}) \leq \frac{\Theta\left(\frac{1}{n}\right)}{\Theta(n^2)} \in O\left(\frac{1}{n^3}\right),$$

as required.

- b. Denote the diameter of G as D . We have $D \in O(n)$ and from Problem 1 we have shown that $\text{vol}(V) \in O(n^2)$.

Recall the variational characterization given for λ_2 as

$$\lambda_2(\mathbf{N}) = \min_{f: V \rightarrow \mathbb{R}, f \neq 0} \frac{\sum_{\{u,v\} \in E} (f(u) - f(v))^2}{\sum_{v \in V} \deg(v) f(v)^2} \quad \text{subject to} \quad \sum_{v \in V} \deg(v) f(v) = 0.$$

For convenience, notice that we could scale any feasible f by a constant and that does not change the variational characterization. Therefore we may replace f by the following function g :

$$g(v) := \frac{f(v)}{\max_{v \in V} f(v) - \min_{v \in V} f(v)}.$$

Denote $M := \max_{v \in V} g(v)$ and $m := \min_{v \in V} g(v)$. By construction we have $M - m = 1$. Take the two vertices u and v corresponding to m and M respectively (so that $g(u) = m$ and $g(v) = M$). By definition, the shortest path connecting them has length $D' \leq D \in O(n)$. Denote this path as $w_1(=u), w_2, \dots, w_{D'-1}, w_{D'}(=v)$.

To bound the numerator, we have, via telescoping sum, that

$$\sum_{i=1}^{D'-1} (g(w_{i+1}) - g(w_i)) = g(v) - g(u) = M - m = 1,$$

so by Cauchy-Schwarz inequality, we have

$$\sum_{i=1}^{D'-1} (g(w_{i+1}) - g(w_i))^2 \geq \frac{\left(\sum_{i=1}^{D'-1} (g(w_{i+1}) - g(w_i))\right)^2}{D' - 1} = \frac{1}{D' - 1} \in \Omega\left(\frac{1}{n}\right).$$

Note that the sum of squared differences over all edges is greater than or equal to the sum of squared differences along this path, i.e.,

$$\sum_{\{u,v\} \in E} (g(u) - g(v))^2 \geq \sum_{i=1}^{D'-1} (g(w_{i+1}) - g(w_i))^2 \in \Omega\left(\frac{1}{n}\right).$$

For the numerator, notice that by $\sum_{v \in V} \deg(v) g(v) = 0$ and $g \neq 0$, we have $m < 0 < M$. Hence $M = 1 + m < 1$ and $m = M - 1 > -1$. As $g(v) \in [m, M]$ for all $v \in V$, we have $0 < g(v)^2 < 1$.

Therefore we have

$$\sum_{v \in V} \deg(v) g(v)^2 < \sum_{v \in V} \deg(v) = \text{vol}(V) \in O(n^2).$$

Finally, we conclude that

$$\lambda_2(\mathbf{N}) = \min_{g: V \rightarrow \mathbb{R}, g \neq 0} \frac{\sum_{\{u,v\} \in E} (g(u) - g(v))^2}{\sum_{v \in V} \deg(v) g(v)^2} \in \frac{\Omega\left(\frac{1}{n}\right)}{O(n^2)} = \Omega\left(\frac{1}{n^3}\right),$$

as required.

Problem 3

- a. Recall that the mixing time is defined by

$$T_{\text{mix}}(G, \varepsilon) := \max_{\mathbf{x}} \min \left\{ t : d_{\text{TV}}(\tilde{\mathbf{W}}^{\top t} \mathbf{x}, \boldsymbol{\pi}) \leq \varepsilon \right\},$$

where $d_{\text{TV}}(\mathbf{x}, \mathbf{y}) = \frac{1}{2} \|\mathbf{x} - \mathbf{y}\|_1$. We just need to show that we can replace $\mathbf{x} \in \mathbb{R}^V$ with $\mathbf{e}_v$ for $v \in V$.

Indeed, as the set of all $\mathbf{e}_v$'s form an orthonormal basis for $\mathbb{R}^V$, we can write any $\mathbf{x}$ as $\mathbf{x} = \sum_{v \in V} x_v \mathbf{e}_v$ where $\sum_{v \in V} x_v = 1$.

By linearity we have

$$\tilde{\mathbf{W}}^{\top t} \mathbf{x} = \tilde{\mathbf{W}}^{\top t} \sum_{v \in V} x_v \mathbf{e}_v = \sum_{v \in V} x_v \tilde{\mathbf{W}}^{\top t} \mathbf{e}_v.$$

Therefore we have

$$\begin{aligned} d_{\text{TV}}(\tilde{\mathbf{W}}^{\top t} \mathbf{x}, \boldsymbol{\pi}) &= d_{\text{TV}}\left(\sum_{v \in V} x_v \tilde{\mathbf{W}}^{\top t} \mathbf{e}_v, \sum_{v \in V} x_v \boldsymbol{\pi}\right) && \text{(by } \sum_{v \in V} x_v = 1\text{)} \\ &= \frac{1}{2} \left\| \sum_{v \in V} x_v \tilde{\mathbf{W}}^{\top t} \mathbf{e}_v - \sum_{v \in V} x_v \boldsymbol{\pi} \right\|_1 && \text{(by definition)} \\ &\leq \frac{1}{2} \sum_{v \in V} x_v \left\| \tilde{\mathbf{W}}^{\top t} \mathbf{e}_v - \boldsymbol{\pi} \right\|_1 && \text{(triangle inequality)} \\ &= \sum_{v \in V} x_v d_{\text{TV}}(\tilde{\mathbf{W}}^{\top t} \mathbf{e}_v, \boldsymbol{\pi}) && \text{(by definition)} \\ &\leq \left(\sum_{v \in V} x_v \right) \left(\max_{v \in V} d_{\text{TV}}(\tilde{\mathbf{W}}^{\top t} \mathbf{e}_v, \boldsymbol{\pi}) \right) && \text{(splitting terms)} \\ &= \max_{v \in V} d_{\text{TV}}(\tilde{\mathbf{W}}^{\top t} \mathbf{e}_v, \boldsymbol{\pi}). && \text{(by } \sum_{v \in V} x_v = 1\text{)} \end{aligned}$$

Therefore taking the maximum over all $\mathbf{x} \in \mathbb{R}^V$ is equivalent to taking the maximum over all $v \in V$ after replacing $\mathbf{x}$ with $\mathbf{e}_v$. Hence the definition remains unchanged.

Problem 4

- a. As $\mathbf{A}$ is symmetric, we could use the variational characterization for its largest eigenvalue as

$$\lambda_{\max}(\mathbf{A}) = \max_{\mathbf{x} \in \mathbb{R}^n \setminus \{\mathbf{0}\}} R_{\mathbf{A}}(\mathbf{x}) = \max_{\mathbf{x} \in \mathbb{R}^n \setminus \{\mathbf{0}\}} \frac{\mathbf{x}^{\top} \mathbf{A} \mathbf{x}}{\mathbf{x}^{\top} \mathbf{x}}.$$

Notice that the numerator equals $\sum_{u \in V} \sum_{v \in V} \mathbf{x}_u \mathbf{A}_{uv} \mathbf{x}_v$. Take $\mathbf{x}$ to be the indicator vector of V , so that $\mathbf{x}_v = 1$ when $v \in V$ and 0 otherwise. As $\mathbf{A}_{uv} = 1$ when $\{u, v\} \in E$ and 0 otherwise, this is equivalent to $\sum_{\{u, v\} \in E} \mathbf{x}_u \mathbf{x}_v = 2|E|$, as each edge

is counted twice. On the other hand, the denominator equals $\sum_{v \in V} \mathbf{x}_v^2 = |V|$.

For an induced subgraph $G[W]$, we take the respective indicator vector $\mathbf{x}_W$. By taking the maximum over all induced subgraphs, we have

$$\lambda_{\max}(\mathbf{A}) = 2 \cdot \max_{W \subseteq V} \frac{|E(G[W])|}{|W|}.$$

Now for any (not necessarily induced) non-null subgraph H , consider the corresponding induced subgraph $G[V(H)]$. By definition we have $E(H) \subseteq E(G[V(H)])$, so $|E(G[V(H)])| \geq |E(H)|$. Therefore

$$\lambda_{\max}(\mathbf{A}) = 2 \cdot \max_{W \subseteq V} \frac{|E(G[W])|}{|W|} \geq 2 \cdot \max_{H \subseteq G} \frac{|E(H)|}{|V(H)|},$$

as required.